

10: People: Cognitive factors; Designing for Inclusion

IN3065/INM355 | 2025–2026
User-Centred System Design

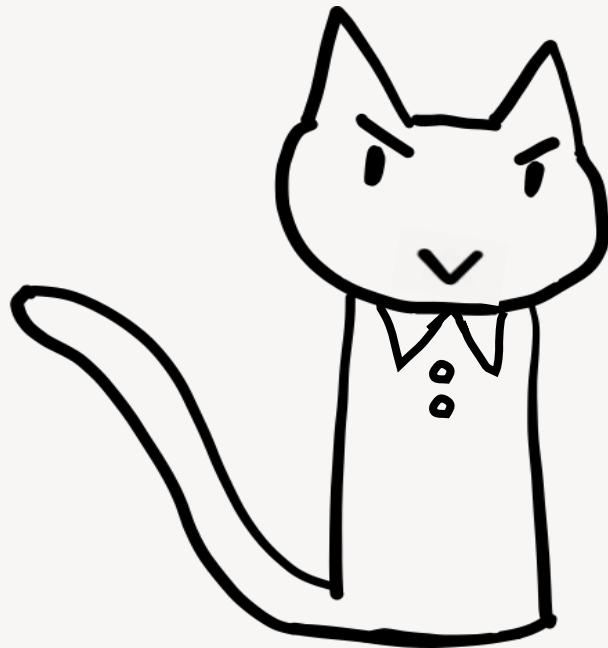
Dr Tracey Booth (she/her)
Centre for HCI Design | Computer Science | SST
tracey.booth.2@citystgeorges.ac.uk

Module Evaluation Questionnaire

Please fill this in ASAP

<https://city.surveys.evasysplus.co.uk>

Closes Sunday 12th April

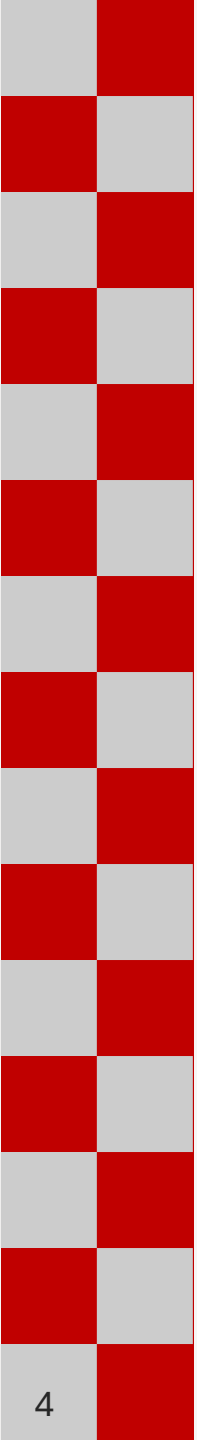


Session 10: Learning objectives



By completing this session, including any practical activity and assigned readings, you should be able to:

- Consider characteristics of human **perception** when designing interfaces.
- Summarise key features of **cognition**: memory, attention, cognitive load, and discuss important implications of these features for the design of interfaces and interactions.
- Explain why designing for **inclusion** is important.
- Describe some considerations and techniques when designing for users with **different abilities/disabilities** and from **different cultures**.

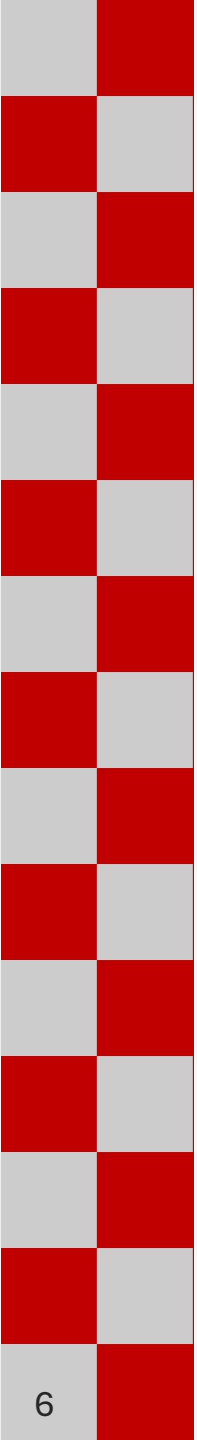


Understanding People

Why might we want to **understand people**?

If we understand people, we can design systems for them that:

- Exploit what they do well.
- Compensate for weakness.
- Enhance abilities.
- Make difficult or impossible things easier or possible.
- Account for reducing or failing capability.



Perception & Cognition

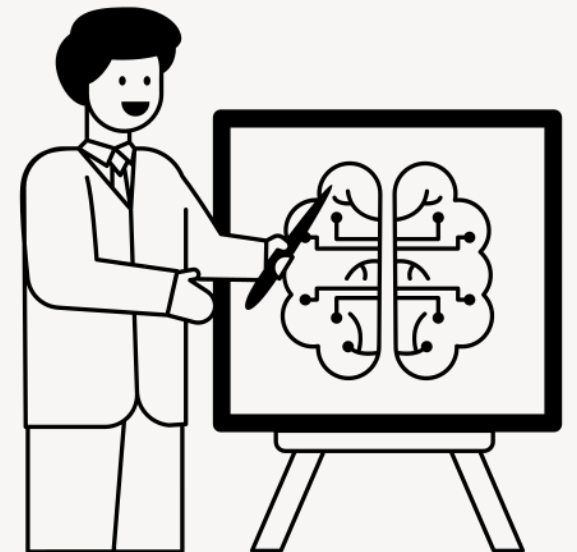
What is **human cognition**?

“Cognition includes all forms of knowing and awareness, such as perceiving, conceiving, remembering, reasoning, judging, imagining, and problem solving”

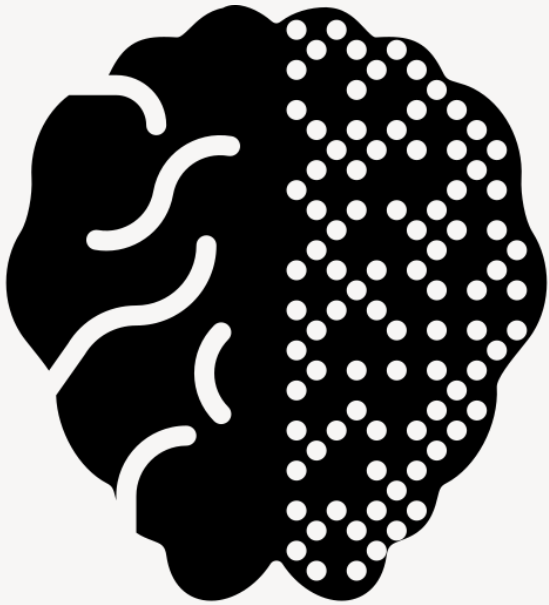
— (APA Dictionary of Psychology, 2018)

“All processes by which the sensory input is transformed, reduced, elaborated, stored, recovered, and used.”

— (Ulrich Neisser, 1967)

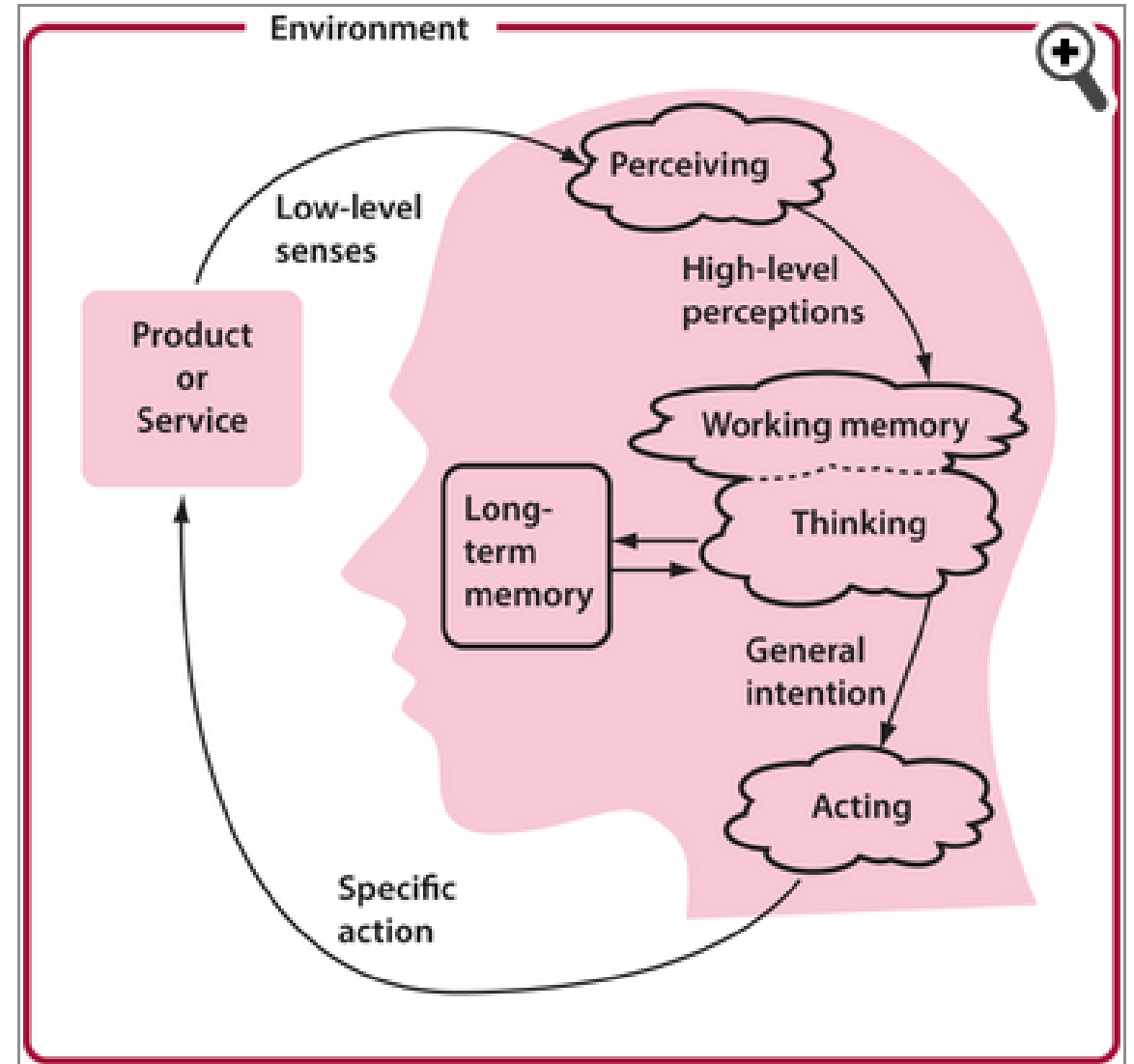


Relevance of theories of cognition to IxD



- To help us understand and explain what is happening in a given situation.
- Theories of cognition can help when we are designing interactive technologies, by:
 - Providing knowledge about what users can and cannot be expected to do (**predictive uses**).
 - Identifying and explaining the nature and causes of the problems users encounter (**explanatory uses**).
 - Applying techniques to help design more effective interaction (**generative uses**).

Cognitive processes and HCI



From <http://www.inclusivedesigntoolkit.com/UCthinking/thinking.html>

Cognitive processes **relevant for design**

- **Perception**
- **Attention**
- **Memory**
- Learning
- Reading, speaking and listening
- Problem-solving, planning, reasoning and decision-making....

In this module we will look briefly at just a few concepts (the ones in red above) that are particularly relevant, but for more information, see:

- Benyon (2019) Designing User Experience (Ch21, Ch22, Ch23, Ch25)
- Sharp, Rogers & Preece (2023) Interaction Design textbook Ch4
- Johnson (2021) Designing with the mind in mind



Human cognition

Perception

Perception

- Involves the five senses
 - Hearing (**auditory** channel)
 - Vision (**visual** channel)
 - Touch (**haptic** channel)
 - Smell (**olfaction**)
 - Taste (**gustation**)
- Also involves perception of space and time.



See also Benyon (2019) [Perception and Navigation](#)
in *Designing User Experience 4th ed.* (Ch 25, pp589-608)

Hearing

- Hearing involves sensing and processing **soundwaves**—complex waveforms composed of many frequencies.
- We can determine the **direction** and **distance** of sound sources.
- Human hearing can be **selective**, and is very good at distinguishing speech from noise.
- Human hearing range: ~20Hz – 20,000Hz
- Hearing **decreases with age**.



Your ears are

LEFT



RIGHT



43

years old

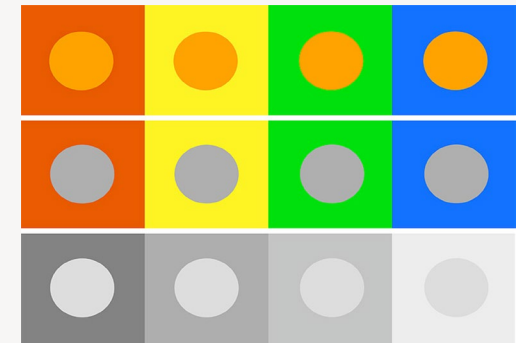
<https://www.youtube.com/watch?v=iN3PBpInNJM&t=54s>

13,904

Frequency Hz

Vision

- Involves receiving an external stimulus (light), processing and interpreting it in terms of objects, depth, colour, etc.
- How well we see something depends on ambient lighting, image brightness, image and image complexity.
- Tuned to recognition of movement and boundaries
- Also tuned for faces
- Highly directional—about 60° for each eye:
 - *Central* vision is good for detail
 - *Peripheral* vision is good for movement, but poor for details
- **Acuity** (ability to see complexity/details) increases with **brightness** and **contrast**

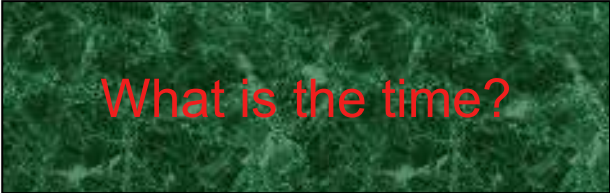


Vision: Colour

- Colour is described in terms of **hue**, **saturation** and **lightness/brightness**.
- Colour sensitivity varies between individuals and between colours.
 - Most people have better acuity **near the middle of spectrum** (e.g. yellow)
- Reaction to colour is **subjective** and **culture dependent**.



Which is easiest to read and why?



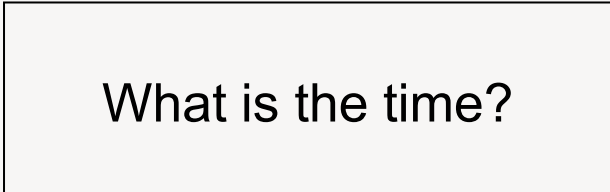
What is the time?



What is the time?



What is the time?



What is the time?



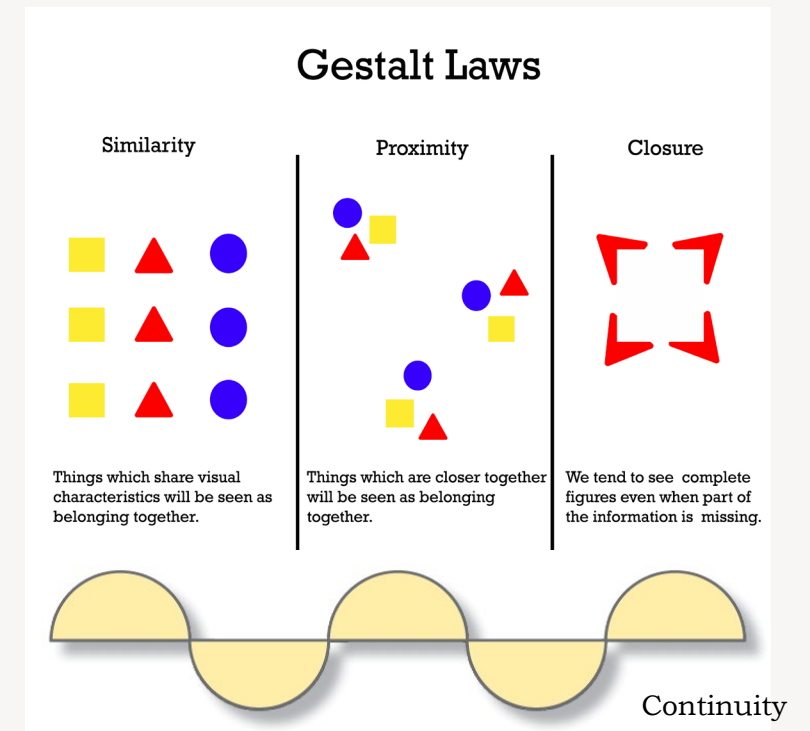
What is the time?

- Adapted from Interaction Design book slides, <http://www.id-book.com/>
- Also see <https://www.inclusivedesigntoolkit.com/UCvision/vision.html>

Gestalt laws of perception

Some most relevant for interface design:

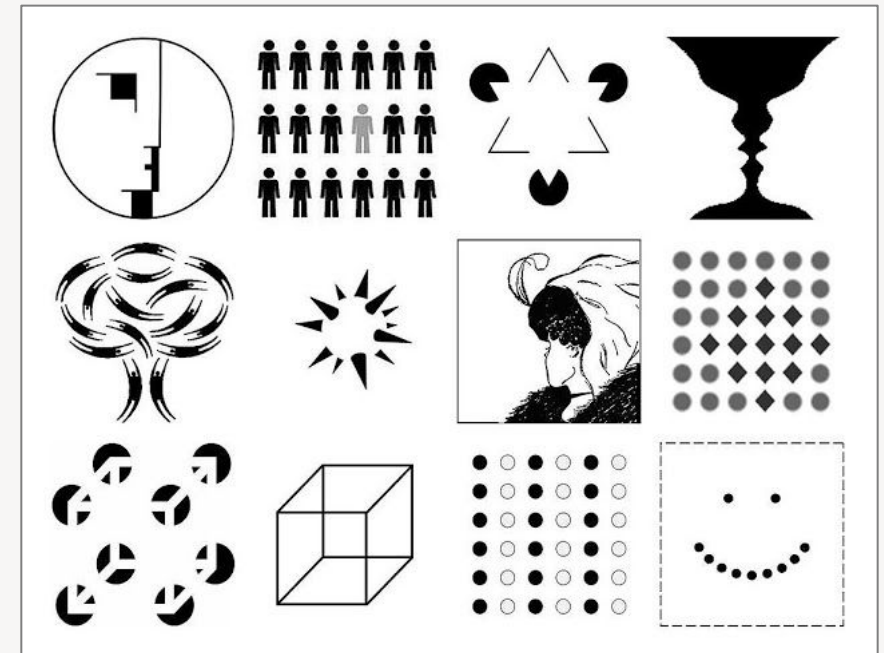
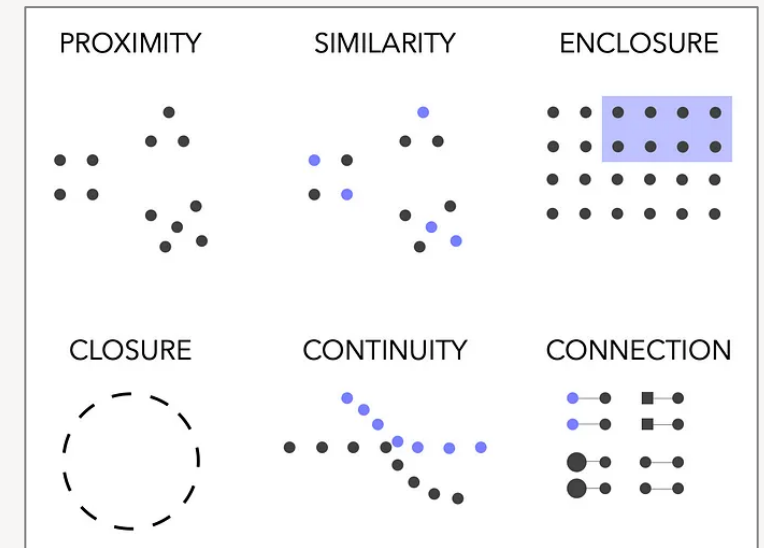
- **Similarity**: similar objects tend to be grouped together.
- **Proximity**: things close in space or time will be perceived together.
- **Closure**: closed figures are perceived more easily; users tend to supply missing information.
- **Continuity**: people tend to perceive smooth, continuous patterns.
- **Pragnanz**: people perceive complex figures in simplest, clearest, most regular form.
- **Common region**: objects in a bounded area are perceived as a group.



Design implications

Perception and Gestalt laws

- Avoid visual noise and clutter.
- Provide visual structure.
- Consider visual flow.
- Be consistent.
- Colour is useful but don't rely on it alone.
- Use of audio is good for alerts but take context into account.



Questions?





Human cognition

Memory

Memory: 3 types

It has been suggested that people have three types of memory:

- **Sensory memory** – specific to each of the senses; holds information for a very brief period of time.
- **Working memory** - limited capacity memory for current information; rapid access and decay.
- **Long-term memory** - main store of information - our usual sense of memory; unlimited capacity; persistent.

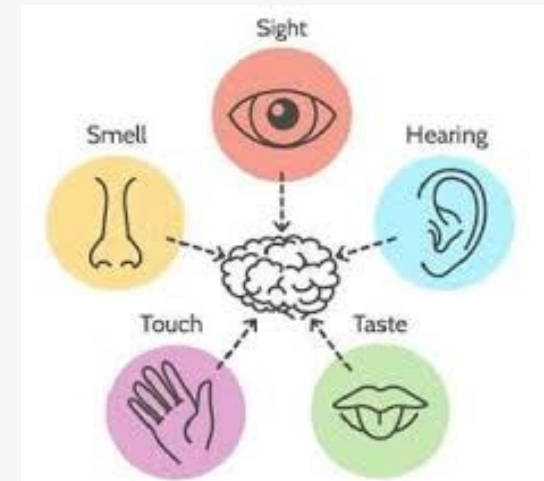


Working Memory = 5-20

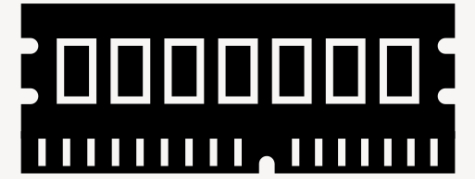


Sensory memory

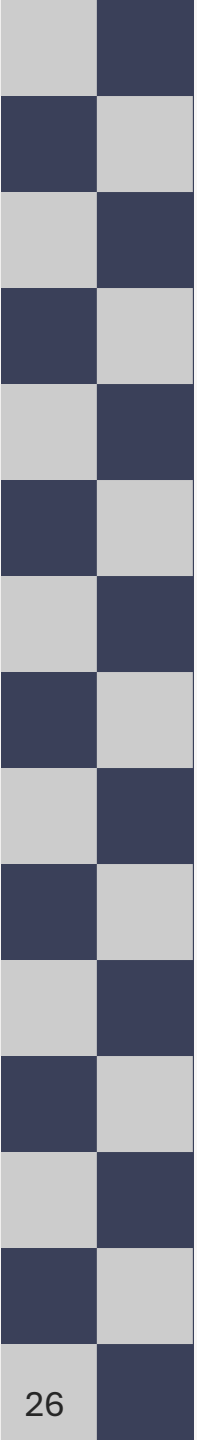
- Sensory memory retains an impression of sensory information for **very short periods of time**.
- Types:
 - **Iconic** (related to vision)
 - **Haptic** (related to the sense of touch)
 - **Echoic** (related to hearing)
 - **Olfactory** (related to smell)
 - **Gustatory** (related to taste)
- Information is **transferred** from sensory memory to working memory when we pay attention to it.
- Information not attended to will be overwritten and **lost**.



Working memory



- Research has shown that memory over short periods of time (working memory) has different characteristics to memory over longer periods.
- Working memory provides a temporary store of facts (sometimes called **short term memory**).
- **Fast** to store and retrieve information.
- But information **rapidly decays** or gets **overwritten** with new input.
- In general, people remember most **recent** information best
 - but this effect is removed if there is interference from another task—working memory can be damaged.



Memory exercises



Hen

Penguin

Camel

Dog

Cat

Monkey

Mouse

Rat

Sheep

Tiger

Horse

Snake

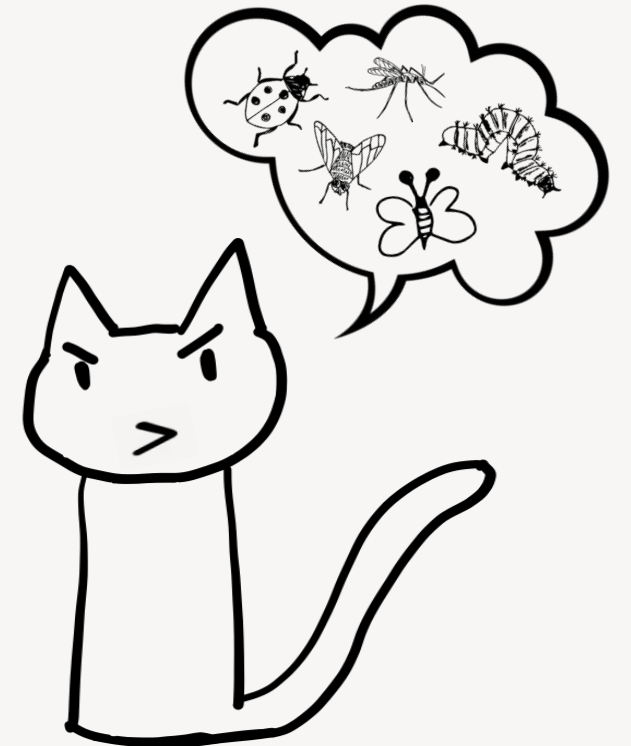
Spider

Elephant

Lion

How many can you remember?

- Write them down
- Count them
- ?/15?
- Did you use any kind of strategy to help you remember?





melca nhe smeou

odg tac greti

tra ersho niol

gipunen klithy phese

 dispre

 tanlehep yemkon

Working memory capacity & chunking

- Working memory holds a very limited amount of information.
- The capacity of working memory is increased by **chunking**:
 - Forming a meaningful whole from individual pieces of information.
 - e.g. the telephone number 020-7040-8000 is easier to remember than the string of digits 02070408000.
- Experiments suggest that its capacity is **7 +/- 2 chunks** of information, where a **chunk** is a meaningful group of smaller items of information.
- We can help users by showing **patterns, categories, groupings, orderings** of information at the user interface.
- The capacity of working memory depends on the level of abstraction of the “chunks” – e.g. numbers, words, models.

Which is **easiest to remember**? Why?

- sdf jrj hkt ghc nfl bvx
 - Nonsense ‘words’. To remember, need to memorise characters & locations.
- hij hiz wom woq riv kca
 - Can at least say them, even if they’re not real words.
- pat sat rat hat mat fat
 - Real words, and they rhyme.
- the quick brown fox jumped over the lazy dog
 - Sentence, also some will already know it—can be a single chunk.
- 469268731056
 - Just a string of 12 digits; likely need to chunk.
- 020-735-8888
 - Structured; easier to remember than string of digits.



Working memory: **implications for design**

Reduce working memory load

- Don't expect users to remember lots of temporary facts
- Don't expect users to be able to deal with lots of different information arriving simultaneously from different sensory channels.
- You can **help users chunk** information in working memory by presenting data in ways that make patterns clear - sort, order, group information.

Plan a journey

[New](#) [My Journeys](#) [Recents](#)

Leaving: now [change time](#) >

[Plan my journey](#)


Live arrivals



Maps



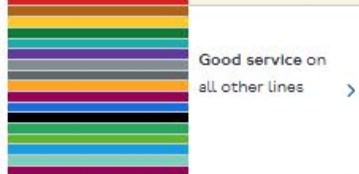
Nearby

Journey Planner launched in 2002

See more of TfL's key improvements over the last 25 years

Tube, DLR, London Overground, Elizabeth line and Tram

Piccadilly	Part closure Minor delays	>
Central	Minor delays	>
Hammersmith & City	Minor delays	>
Windrush	Minor delays	>



Good service on all other lines >

This weekend >

View all statuses >

Buses +

Traffic +

River +

IFS Cloud Cable Car +



Enter the TfL 25 festive prize draw

More than 25 amazing prizes up for grabs

TfL Go

Our live Tube map app helps you plan travel on the move

Maps

Maps and guides to help you get around



Top up Oyster



Congestion Charge



Santander Cycles



IFS Cloud Cable Car



Major works and events

Find out about works and events in London which may affect your journey

[Report a crime or incident](#)
[Sustainability](#)
[Check the best times to travel](#)
[TfL Go app](#)
[Experience London blog](#)

When 7+/-2 is **not** appropriate

The limited capacity of working memory is not relevant when thinking about:

- The number of items in an on-screen menu.
- Or icons in a toolbar.
- Or tabs on a web page.

Why not?

See also: <https://uxmyths.com/post/931925744/myth-23-choices-should-always-be-limited-to-seven>

Long-term memory

- Our ‘permanent’ store of information - our everyday sense of memory.
- Consists of inter-connected chunks of information (associations).
- Capacity nearly infinite - storing is always possible, the problem is remembering.
- Slow, but variable access time.
- Little decay.
- Performance decreases with age, but the store becomes richer.
- Different types of memory: episodic, semantic etc.
- Main processes:
 - **Storing (memorising)**—through rehearsal or repeated exposure.
 - **Retrieval**—recognition versus recall
- **Frequency** and **recency** effects: the more you use a memory, and the more recently you have used it, the easier it is to recall

Recognition vs recall

- Two main ways of thinking about retrieval
 - **Recognition** involves identifying an item when presented with it.
 - **Recall** involves remembering the item using a different cue.
- Recognition is
 - the **less complex** cognitive activity.
 - typically, **faster** than recall.
 - **less prone to error**.

Who is the British Prime Minister?

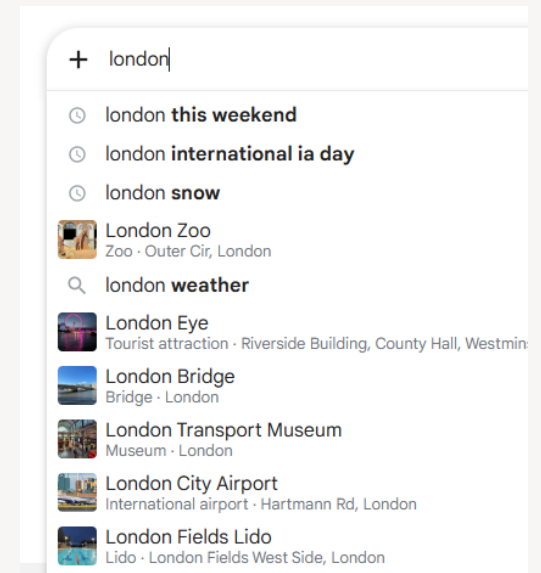
- a) Theresa May
- b) Tony Blair
- c) David Cameron
- d) Rishi Sunak
- e) Keir Starmer
- f) Boris Johnson



- ? Is this recognition or recall?
- ? Why is this relevant for interaction design?

Long-term memory: **implications for design**

- Most gesture-based, voice-based and command-based interactions require users to recall them from a memorised set; GUIs require recognition of visual items.
- Think about designs that **require recognition rather than recall** of information.
 - Present options on screens, are allowing users to recognise them.
 - Lists of visited URLs, recent searches, recently used files, predictive text, autocomplete-enabled fields etc, all aid recognition.
- We can **help people memorise things**:
 - It is easier to memorise meaningful information:
 - Structure information to make it meaningful and help chunking—use categories, ordering, associations...
- Support the user in **offloading information** to “external memory”.



Questions?





Human cognition

Attention

Attention

- Attention is about **focusing** the mind on one of several simultaneous possible thoughts.
- Attention is **selective**—we can only focus on one thing at a time.
- Can still do background tasks, or time-slice between tasks (multi-tasking).
- Attention switching can be **voluntary**, or **involuntary**.
- Complex tasks hold attention better than mundane ones.
- Humans are poor at monitoring tasks—error prone.
- Sensory input disrupts focused attention—hard to ignore



[https://www.youtube.com/
watch?v=IGQmdoK_ZfY&t=
5s](https://www.youtube.com/watch?v=IGQmdoK_ZfY&t=5s)



(c) 2010 Daniel J. Simons

Attention: (some) implications for design

- Guide attention to relevant information, e.g. use colour, movement, alerts, structure to direct attention.
- Avoid distractions.
- Avoid clutter.
- Engage users in active tasks, not just monitoring.
- Provide breakpoints in lengthy tasks.
- See also:
 - <http://www.inclusivedesign toolkit.com/UCthinking/thinking.html#attention>
 - Benyon (2019) 21.3. Attention. In *Designing User Experience* (pp512-521)

Questions?





Human cognition: **Cognitive Load**

What is **cognitive load**

“The cognitive load involved in a task is the cognitive effort (or amount of information processing) required by a person to perform this task.” (Reif, 2010)

- **Intrinsic cognitive load:**
 - The inherent difficulty of a task.
- **Extraneous cognitive load:**
 - Unnecessary mental effort caused by extraneous factors.
- **Germane cognitive load**
 - Mental effort to learn and understand.



Warning: Poor design can unnecessarily *increase* cognitive load.

As designers, we ideally want to minimise extraneous cognitive load, manage intrinsic cognitive load, and optimise germane cognitive load

What **increases** cognitive load?

- Task complexity
- Unfamiliarity / Limited prior knowledge
- Distraction and interruption
- Visual clutter; confusing or over-complicated design.
- Split attention / multi-tasking.
- Stress / fatigue.



How to **reduce** cognitive load

- Keep things **simple**.
- Eliminate **unnecessary steps**
- Be **consistent** and **predictable**.
- **Reuse** common user interface design patterns
- **Offload** tasks to the system.
- Avoid/remove **clutter** that competes for attention
- **Chunk** content.
- **Recognition** rather than recall.
- Redesign with **care**

MINIMISE



Extraneous
Load

MANAGE



Intrinsic
Load

Minimising cognitive load

See also:

- <https://lawsofux.com/articles/2015/design-principles-for-reducing-cognitive-load/>
- <https://boagworld.com/design/cognitive-load/>
- <https://medium.com/design-signals/cognitive-psychology-in-ux-minimising-the-cognitive-load-d97ad8e3115b>
- <https://www.nngroup.com/articles/minimize-cognitive-load/>

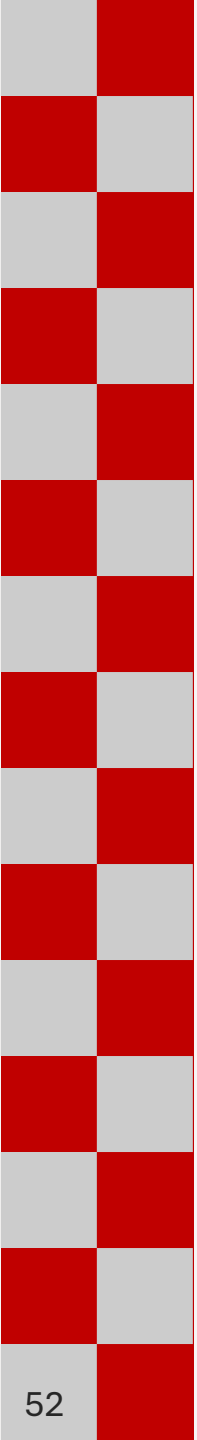
What **users** really, really **want**

- Users want to concentrate on what they are doing (their tasks), not have to think about how to get the system to do the task.
- Psychologically, this makes good sense—humans have limited cognitive capacity and should spend it on the task.
- If the interaction is too complex, the user won't have any capacity left for the task.



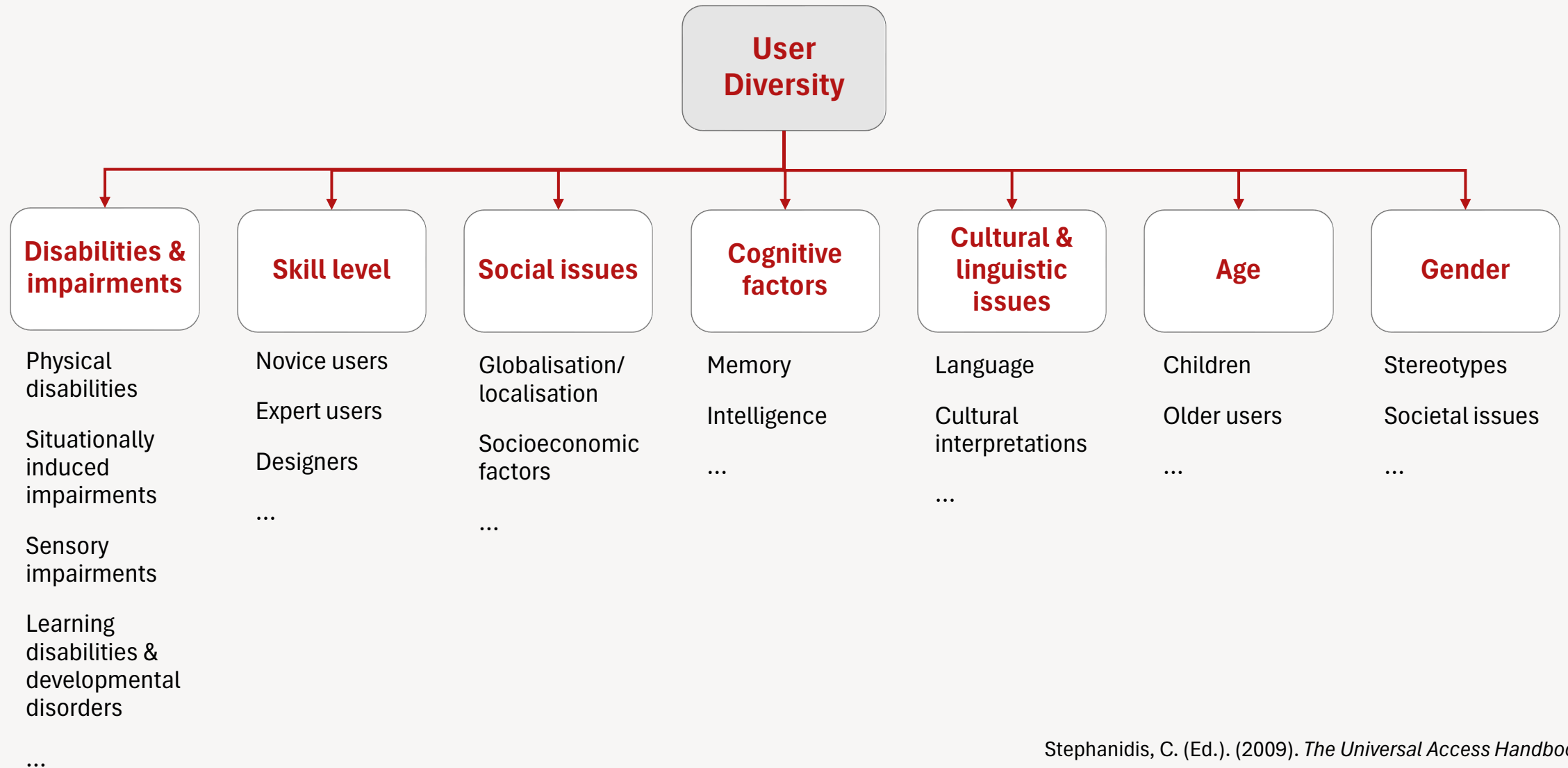
Questions?





Designing for **Inclusion**

Diversity of Users





Designing for Inclusion

Ability / disability

Users with **different abilities**

- **Sensory disabilities**
 - e.g. Vision and hearing impairments
- **Physical disabilities**
 - Impairments to mobility and motor skills
- **Cognitive disabilities**
 - Impairments affecting memory, language and information processing.

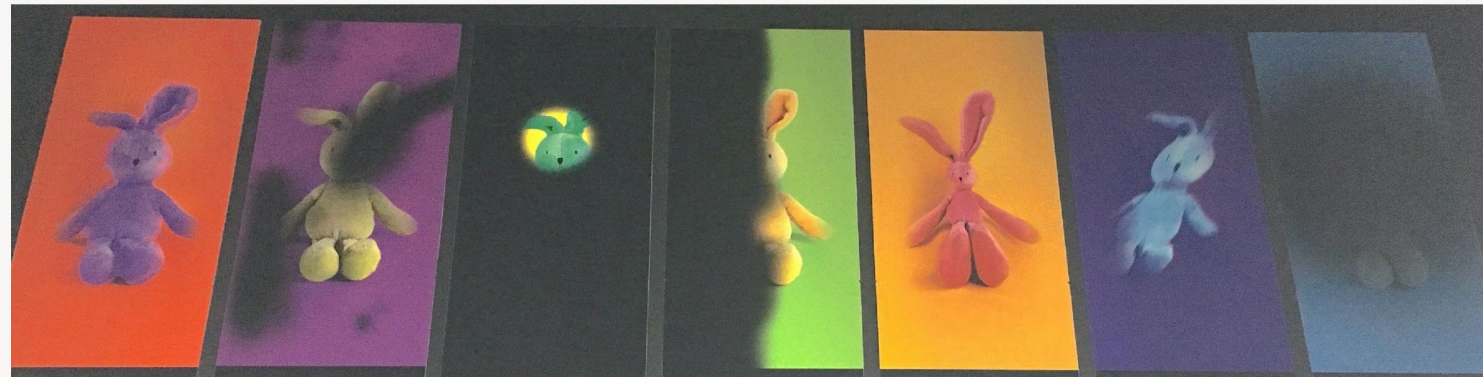


Image: © Rebekah Cupitt 2017

Disability around the **world** and in the **UK**

- Around 1.3 billion people around the world experience significant disability, making up about 16% of the global population—1 in 6 people. ([WHO 2023](#))
- In 2023/24 there were 16.8m people classed as disabled people in the UK—25% of the population ([DWP](#)).
- Impairment and disability become more common with age. The UK population is ageing: 19% were aged 65+ in 2022, projected to rise to 27% by 2072. ([House of Commons Library](#))

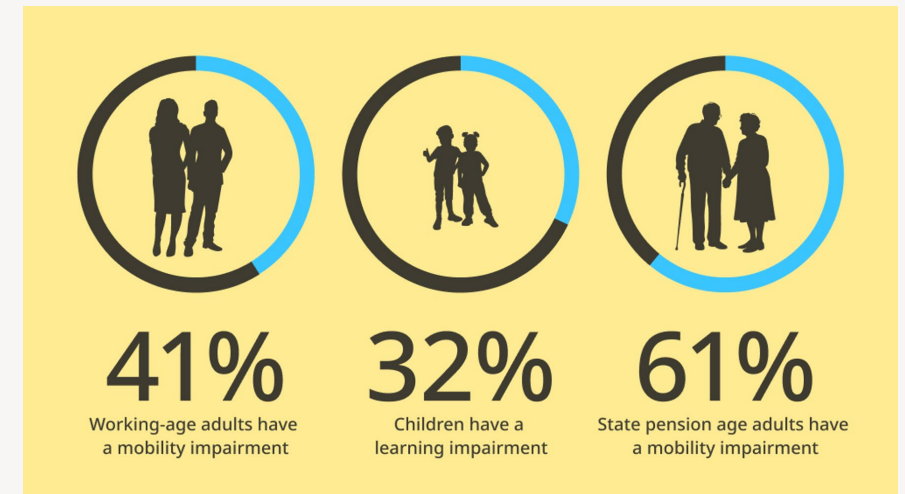


Image: <https://kindera.uk/articles/official-statistics-nearly-a-quarter-of-uk-residents-are-disabled>

Physical disability categories

- Lack of **function** - absence of a limb or paralysis
- Lack of **mobility** - lack of strength, speed, range
- Lack of **muscle coordination** - tremor/uncontrolled movement/lack of accuracy

All of these have implications for interacting with systems



Physical disability examples

- **Paraplegia** – paralysis of lower limbs only
- **Arthritis** – inflammation of one or more joints, which results in pain, swelling, stiffness, and limited movement
- **Peripheral neuropathy** – loss of feeling in finger-tips
- **Cerebral palsy** – neurological condition that affects movement and co-ordination and results in muscle stiffness or floppiness, random and uncontrolled body movements and balance and co-ordination problems

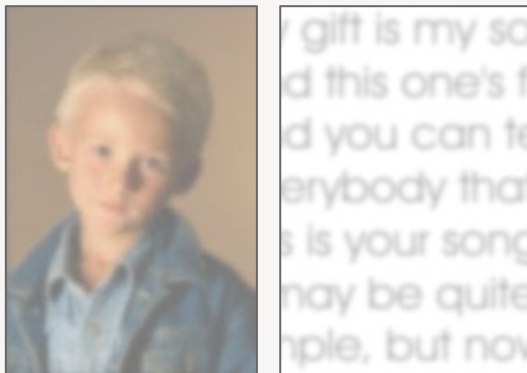
How might these affect interaction with systems or technology?

Visual impairment

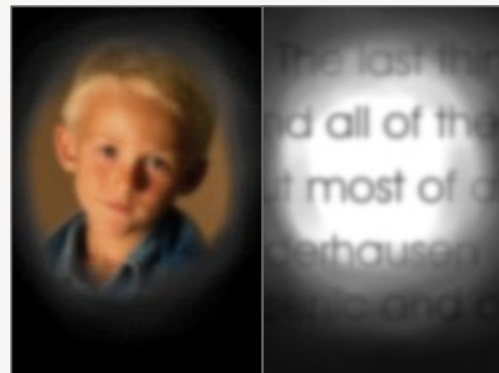
- **Acuity** e.g. blindness, short-sightedness, long-sightedness
- **Visual field** e.g. glaucoma, macular degeneration
- **Colour vision** e.g. red-green colour blindness

<https://coopervision.co.uk/practitioner/clinical-resources/myopia-in-children/myopia-simulator>

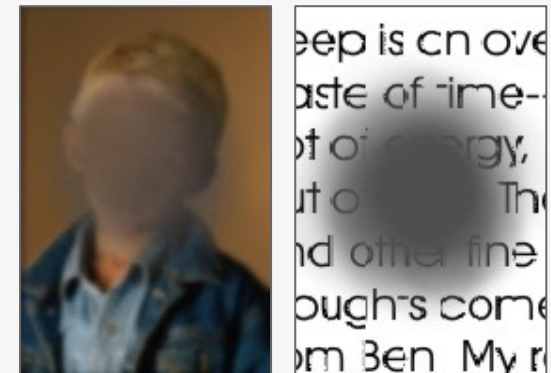
Cataracts



Glaucoma

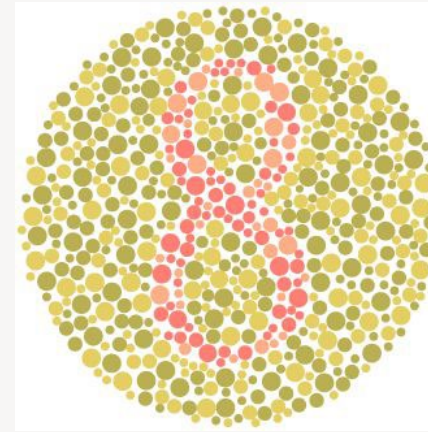


Macular Degeneration

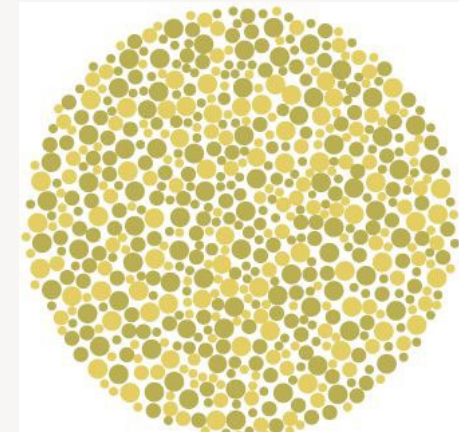


Colour vision

- Approximately 8-10% of the male population and about 0.5% of the female population experience some form of colour deficiency
 - Red-green blindness
 - Blue-yellow blindness (rarer)
- Implications for the use of colour in design—don't rely on colour alone to communicate something.



This is an Ishihara plate commonly used to check for red/green colour blindness.

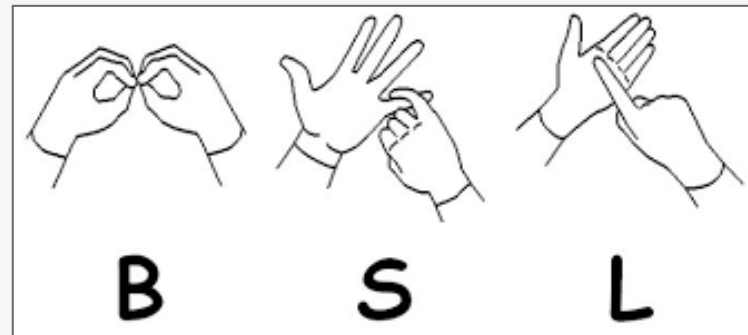


This is what a red/green colour-blind person might see.

See <https://www.colour-blindness.com/colour-blindness-tests/ishihara-colour-test-plates/>

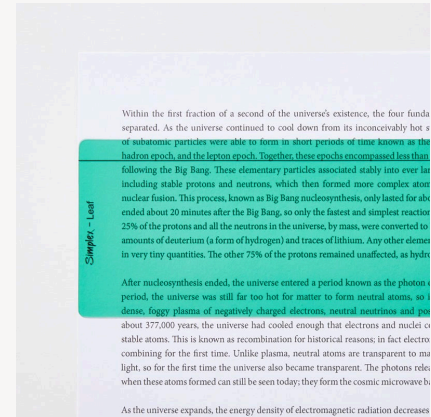
Auditory impairment

- Hearing loss over time
 - 13% of people 65–74 and 26% of those over 75 have hearing impairments.
 - Of persons with serious hearing loss, 55% are over the age of 65
 - Most often a reduction in perceptible (high) frequencies but could also be other causes
- Auditory Hypersensitivity
 - Hyperacusis
 - Phonophobia
- Implications for audible interfaces / feedback.



Cognitive and neurological impairment

- Cognitive functions that may be affected
 - Attention and concentration—e.g. ADHD
 - Memory—e.g. dementia, brain injury
 - Reasoning and problem solving—e.g. brain injury, dementia
 - Language processing, reading and writing—e.g. dyslexia
 - Visuospatial and perceptual skills—e.g. stroke
 - Executive functioning—e.g. ADHD, brain injury
- Related conditions that may also affect interaction
 - Motor coordination—e.g. dyspraxia / DCD, multiple sclerosis
 - Broader developmental or neurological conditions—e.g. autism, stroke, dementia

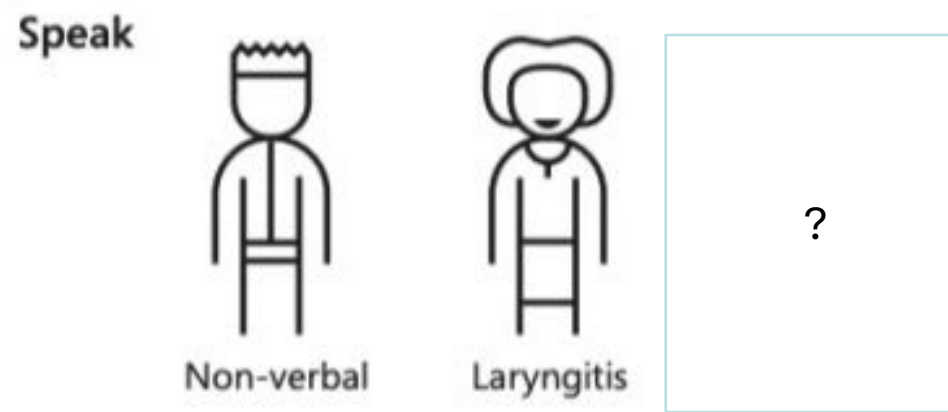
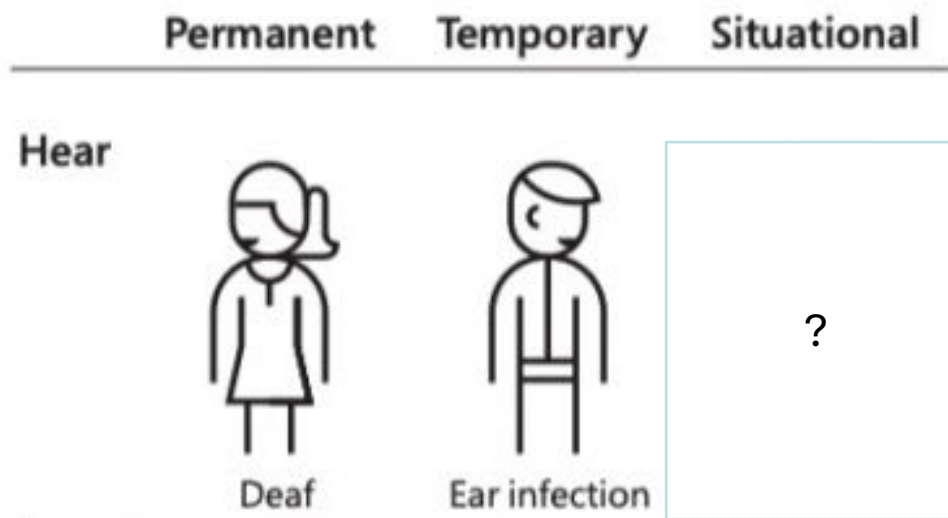
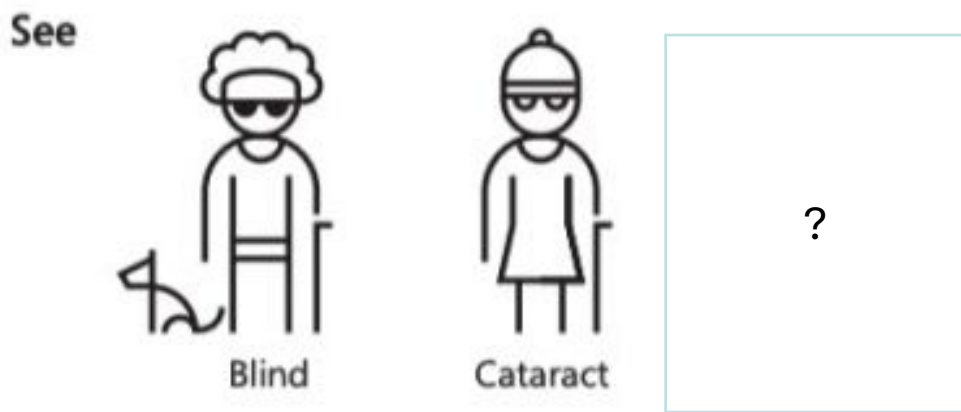
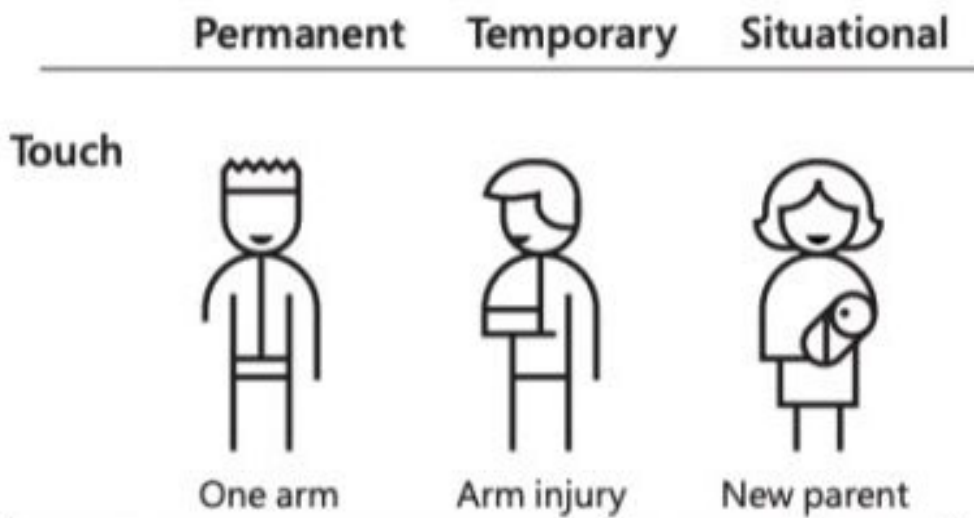


We're only **temporally** non-disabled

Our abilities are situational and also depend on our environment.

- Being
 - in a foreign country,
 - in a bouncing vehicle,
 - in the dark,
 - in a noisy room
- Non-optimal health: Lack of sleep, inebriated, fever
- Injury
- Ageing
- Technological advancement - new products, unfamiliar interfaces.

Persistence of impairments



Orientation to **design**

- By making design decisions, designers have the power to include and exclude.
- *“**Inclusive design** means truly including others in the process. In that spirit, Microsoft partners with companies, non-profits, and communities to explore human-centered design and connect technology to people’s lives in meaningful ways.” (Microsoft)*

<https://www.microsoft.com/design/inclusive/#inclusivethefilm>

Why focus on **inclusion** in design?

- Legal requirements
 - E.g. Equality Act 2010 (<http://www.legislation.gov.uk/ukpga/2010/15/content>)
- Economic arguments
 - Size and value of UK disabled consumer market (>£80 billion).
 - Increase contribution of working disabled people to economy.
- Usability argument
 - Designing for a diverse range of needs can actually appeal to and be usable by more people.
- Moral argument
 - Disabled people have the right to participate in society, that involves having access to mainstream technologies, services, the environment.

[Web] Accessibility

- Providing **access** and **assistance** to people with different **needs**
- Applied to the **environment** but also **technology** and **services**
- **Web accessibility** is the ability of a website or application to be easily navigated, used and understood by a wide range of people, including those who have disabilities.
- It combines web development standards with inclusive design practices to create websites and applications that are accessible to a broad spectrum of people.

Accessibility **standards and guidelines**

- Helpful guides for professionals.
- Ensure consistency between user interfaces.
- Promote the good practice for implementation, design and evaluation.
- Useful for prioritisation of issues.
- Ensure that legal requirements for accessibility are met.



Image: <https://www.blueatlasmarketing.com/resources/a-wcag-digest/>

W3C WAI



- World Wide Web Consortium (**W3C**)
 - The main international standards organization for the World Wide Web
 - Develops international standards for the Web: HTML, CSS, + more
- Web Accessibility Initiative (**WAI**) is responsible for the development of:
 - Resources to understand and achieve web accessibility, including
 - Guidelines for web accessibility which are commonly regarded as the international standard
 - Web Content Accessibility Guidelines (**WCAG**)
 - Accessible Rich Internet Applications (**ARIA**)
- See: <https://www.w3.org/WAI/>
<https://www.w3.org/WAI/standards-guidelines/>

WCAG **POUR** Principles

1. Perceivable

Information and UI components must be presentable to users in ways they can perceive. = people are able to find your content

2. Operable

UI components and navigation must be operable = the interface cannot require interaction that a user cannot perform

3. Understandable

Information and the operation of UI must be understandable = the content or operation cannot be beyond their level of comprehension

4. Robust

Content must be robust enough that it can be interpreted reliably by a wide variety of user agents, including assistive technologies = as technologies and user agents evolve, the content should remain accessible

Questions?





Designing for Inclusion **Culture**

What is “culture”?

“Culture ... is that complex whole which includes knowledge, belief, art, morals, law, custom, and any other capabilities and habits acquired by man as a member of society.”
(Tyler, 1871)

“Culture is a fuzzy set of basic assumptions and values, orientations to life, beliefs, policies, procedures and behavioural conventions that are shared by a group of people, and that influence (but do not determine) each member’s behaviour and his/her interpretations of the ‘meaning’ of other people’s behaviour.”
(Spencer-Oatey, 2008)

“the collective programming of the mind which distinguishes the members of one group or category of people from another.” (Hofstede 1994)

“... the set of attitudes, values, beliefs, and behaviors shared by a group of people, but different for each individual, communicated from one generation to the next”
(Matsumoto, 1996)

What is “culture”?

Broadly speaking, culture might be thought of as the set of shared attitudes, values, goals, conventions and practices, social forms, and material traits of a group or organisation which create patterns of meaning.

BUT

- Culture is a tricky concept... there are also individual variables
- The use of the concept of culture in design depends very much on the area of application and implementation.

Designing for cultural diversity

- Internationalisation

The process of designing and developing, or re-engineering websites, applications, etc, so that they can be used in various *locations*, with minimal changes.

- Localisation

The process of adapting the design to a *culture* — goes beyond translation.



“properly localized” software applications, just like properly localized automobiles, toasters, beverages, and magazines, reflect the values, ethics, morals and language (or languages) of the nation in question”

Designing for cultural diversity: **considerations**

- **Text**

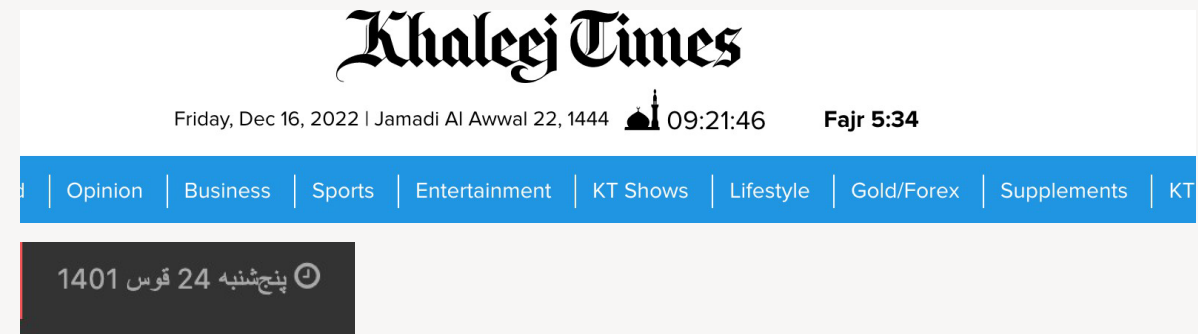
- Should be appropriate for the context, and images should match the message given in the text;
- Important: “correctness”, jargon, non-existent words, character sets, product names, amount of space needed

- **Writing systems**

- How information is displayed will vary according to the system used.
- Consider flow: Left to right and vs versa, top to bottom.
- Is flow still logical?
- Is more space needed?

Designing for cultural diversity: **considerations**

- Dates, time and numbers
 - 03/12/13? 23:45? What year is it? 2022? 1401? 1444?
- Images
 - Meaning and acceptability;
 - Consistency of brand image – the company may want the product to look recognisably similar in different markets, even though images and writing systems may differ.
- Symbols and colours
 - Meaning isn't universal!
(See [Russo & Boor 1993](#))



Adapted from material by Sylwia Frankowska-Takhari

Designing for cultural diversity: **considerations**

- **Regulatory frameworks**

- A locality will have its own framework and may use different systems of measurement. For example, providing the liquid weight in ml or fl oz.

- **Features**

- Features should be relevant and appropriate for the audience/market.
Important to understand the target users and their practices. (see Chavan et al. 2009)

- **Involving users**

- When doing researching with under-researched, minority groups, consider carefully who should be involved and in what role.

Questions?





Alternative approaches to design

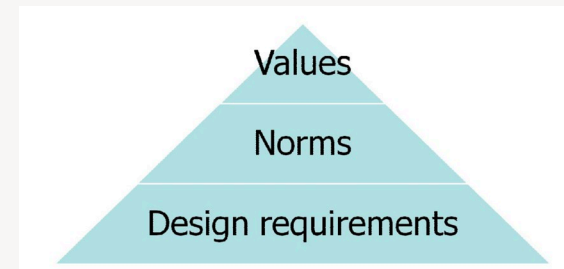
Participatory Design and Co-Design

- Participatory Design emerged from the strong labour movement in Scandinavia in the 1970s.
- These techniques emphasise active involvement of all stakeholders.
- Using techniques such as workplace visits, future workshops, co-operative prototyping, paper prototyping.



Images from Bourazeri, A. & Stumpf, S. (2018). Co-designing smart home technology with people with dementia or Parkinson's disease. NordiCHI '18

Value-Sensitive Design (VSD)



- “Using our moral and technical imaginations to create responsible innovations” (Friedman and Hendry, 2019)
- A “value” is what people consider important in life.
- Involves:
 - **Conceptual investigations** to understand the stakeholders, their values and any values conflicts that might arise for these stakeholders through the use of the technology.
 - **Empirical investigations** (qualitative or quantitative studies) to inform understanding of the users' values, needs, and practices.
 - **Technical investigations** either to analyse how people use related technologies, or to design systems to support values identified in the conceptual and empirical investigations.

Questions?



Session 10 Summary

- Human abilities and limitations have important implications for interaction design and cognitive psychology can give us some insight into these.
- The way that we design interactive systems impacts on how well users can perceive, remember, learn and attend to their tasks.
- Theories of cognition may be brought into design through mechanisms such as high-level design principles.
- Users are diverse, and there are good arguments for designing for inclusion.
- Standards and guidelines (e.g. WCAG) can be used to ensure systems are accessible to a broad range of users, with different abilities/disabilities.
 - This can benefit all users
- Designing for different cultures can be complex—there's a lot to consider and context is key.